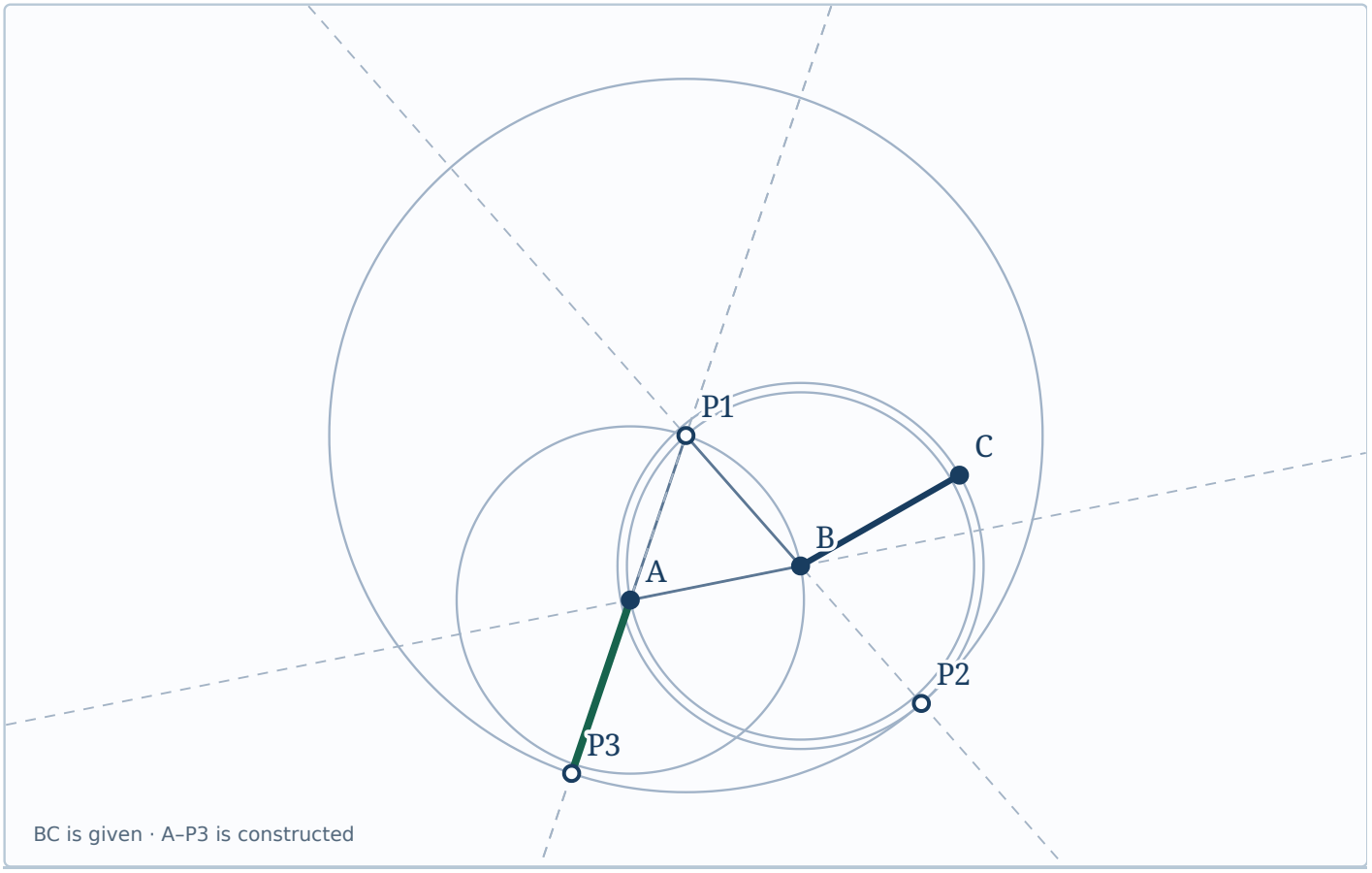
Your I.2 construction

Layout: separate. 11 chosen operations. The drawing and transcript below preserve your actual choices, including detours; they do not substitute a prepared solution.

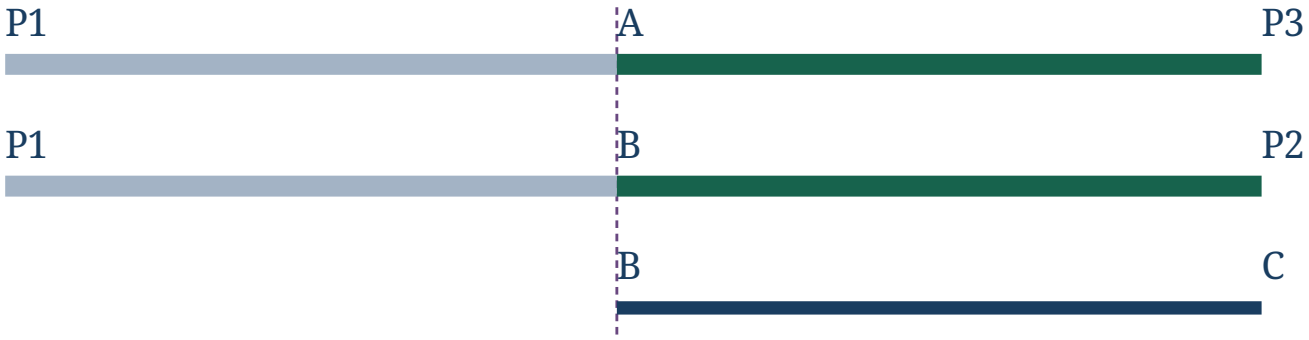
Continue this construction in the app · Save the editable drawing JSON

The drawing you made



The equal parts can now disappear

I.2’s equal-remainder chain is present: A–P3 = B–C.



Grey: equal parts removed. Green: equal remainders. Blue: the given reference.

1. P1–A = P1–B

   The two circles on A–B supply the equilateral triangle (I.1).

2. B–P2 = B–C

   Both are radii of c5, centred at B.

3. P1–P3 = P1–P2

   Both are radii of c7, centred at P1.

4. A–P3 = B–P2 = B–C

   P3 is beyond A and P2 beyond B. Remove equal parts P1–A and P1–B from the equal wholes: common notion 3, then common notion 1.

Match your labels to the printed I.2 argument

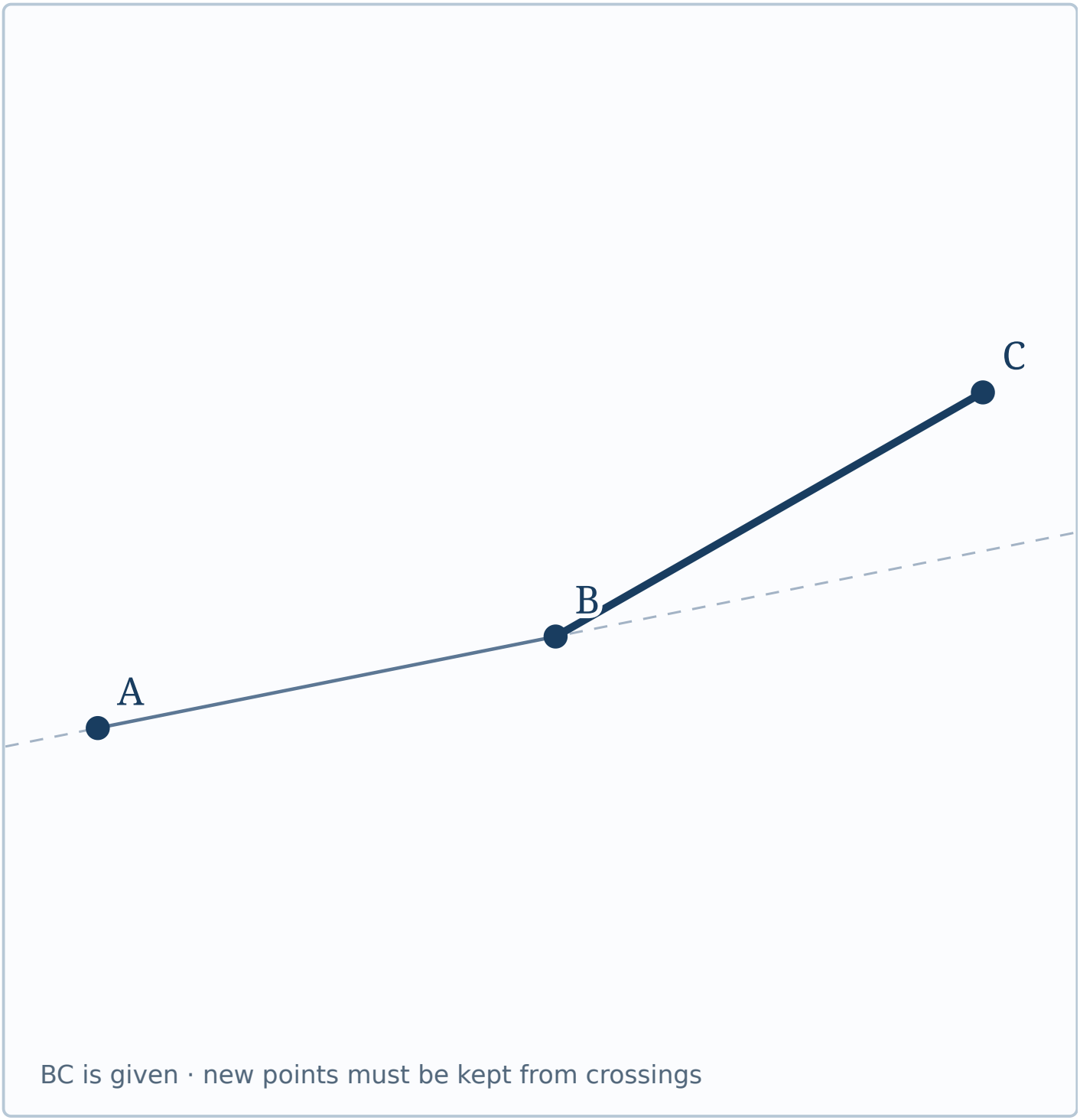
This drawing	Source	Role
A	A	Given point
B	B	Reference anchor
C	Γ	Reference endpoint
P1	Δ	Equilateral vertex
P2	H	Beyond the anchor
P3	Λ	Copied endpoint

This recognizes construction dependencies and the required point order in a floating-point model of the real plane. It is not a formal proof checker; existence of intersections is an additional assumption in Euclid’s presentation.

Replay your operations

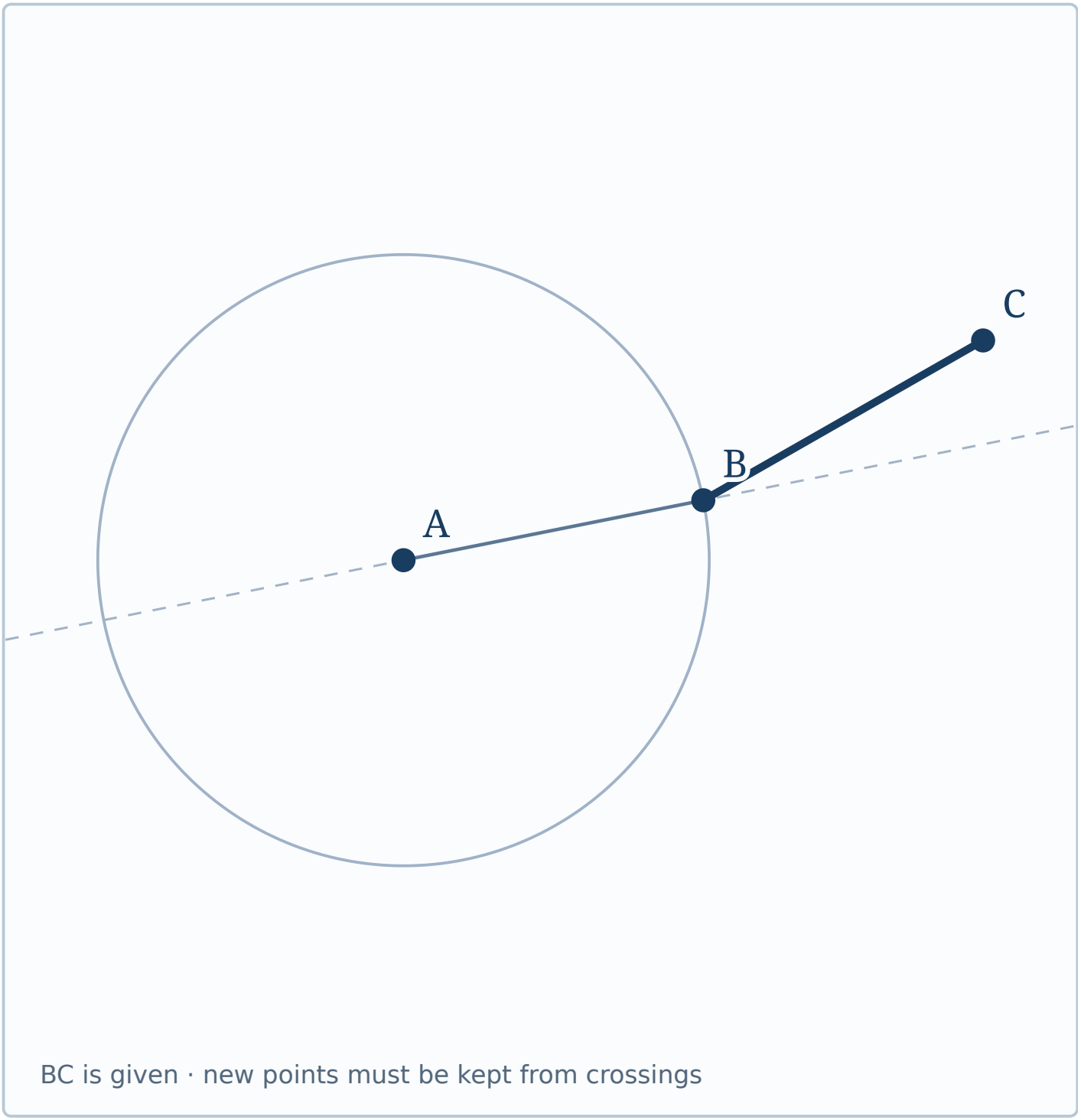
Each diagram contains only the points and curves available at that operation. A kept crossing records its two parent curves and branch. Curves are named c or l plus their creation number; points are numbered in order of keeping.

1. Join and continue A–B



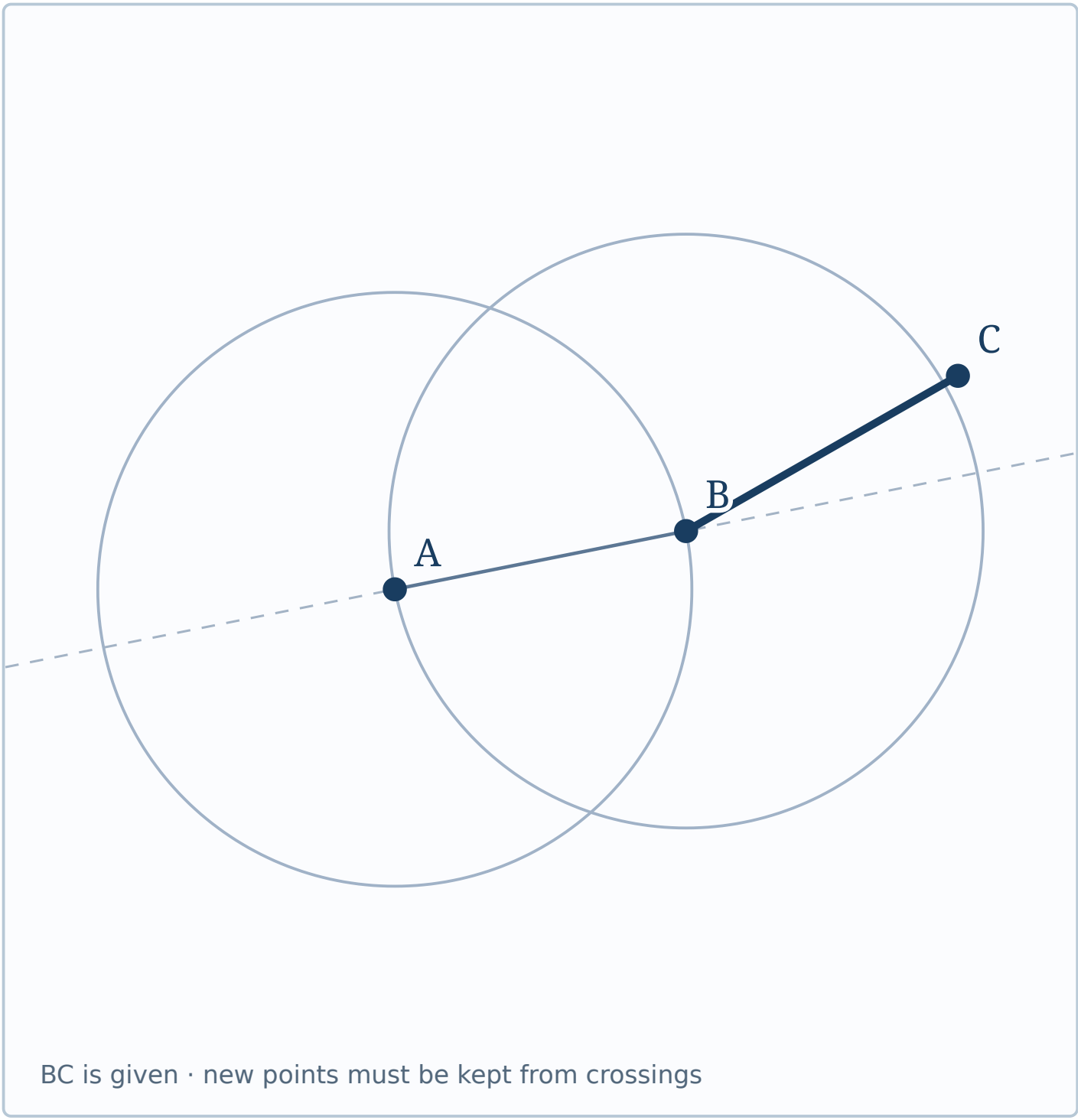
BC is given · new points must be kept from crossings

2. Circle centred at A, through B



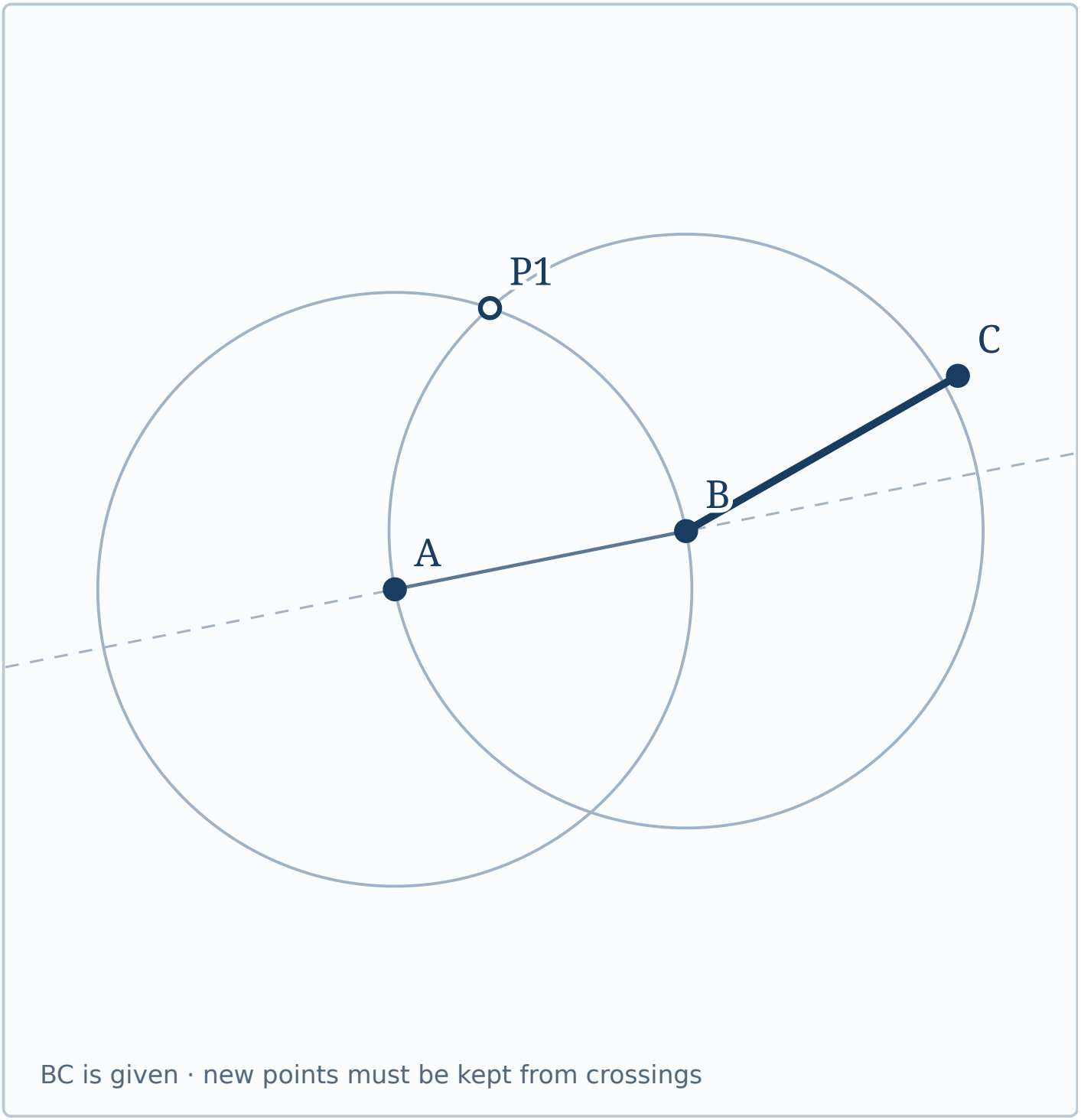
BC is given · new points must be kept from crossings

3. Circle centred at B, through A



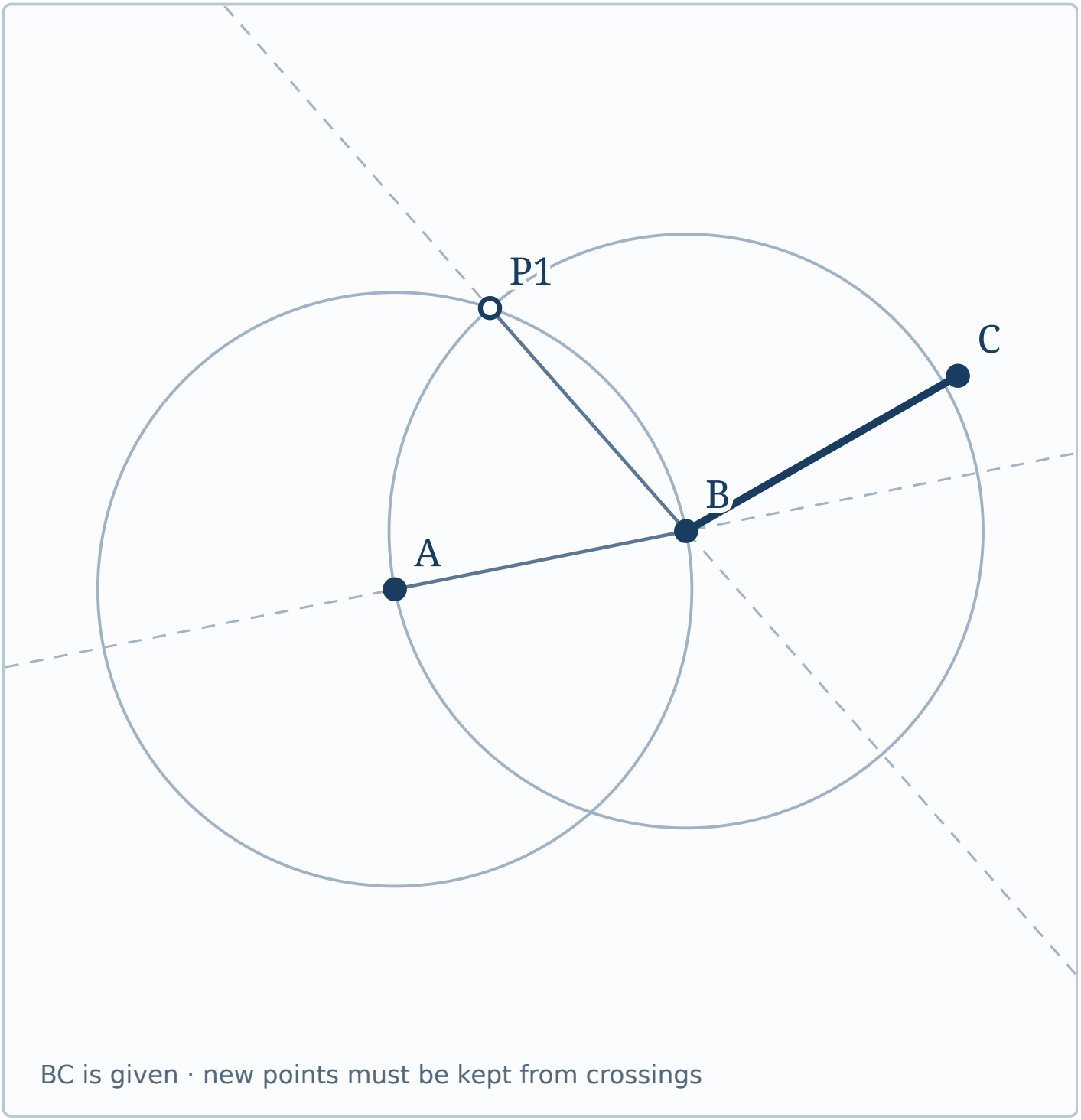
BC is given · new points must be kept from crossings

4. Keep crossing 1 of c2 and c3

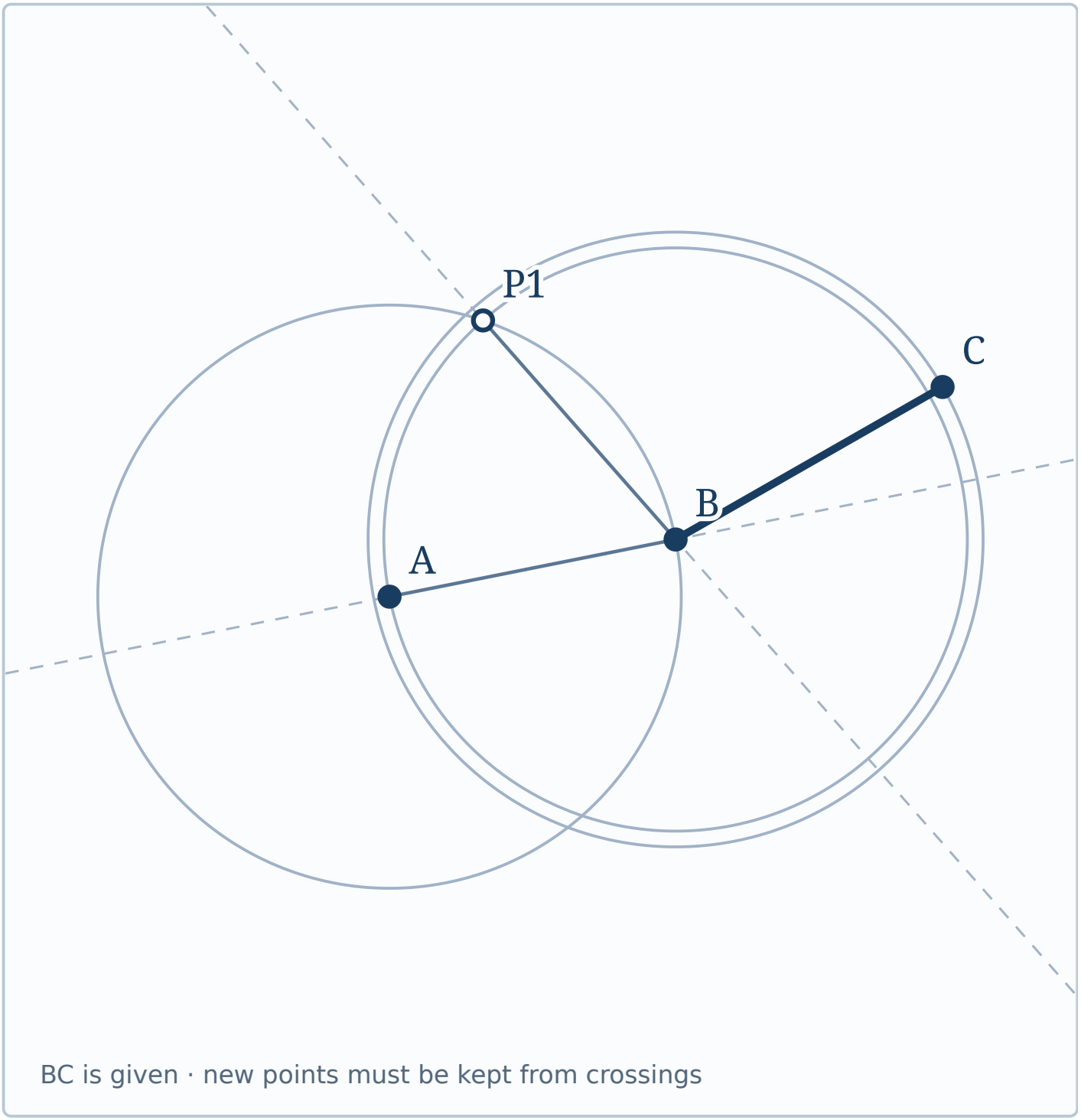


BC is given · new points must be kept from crossings

5. Join and continue P1–B

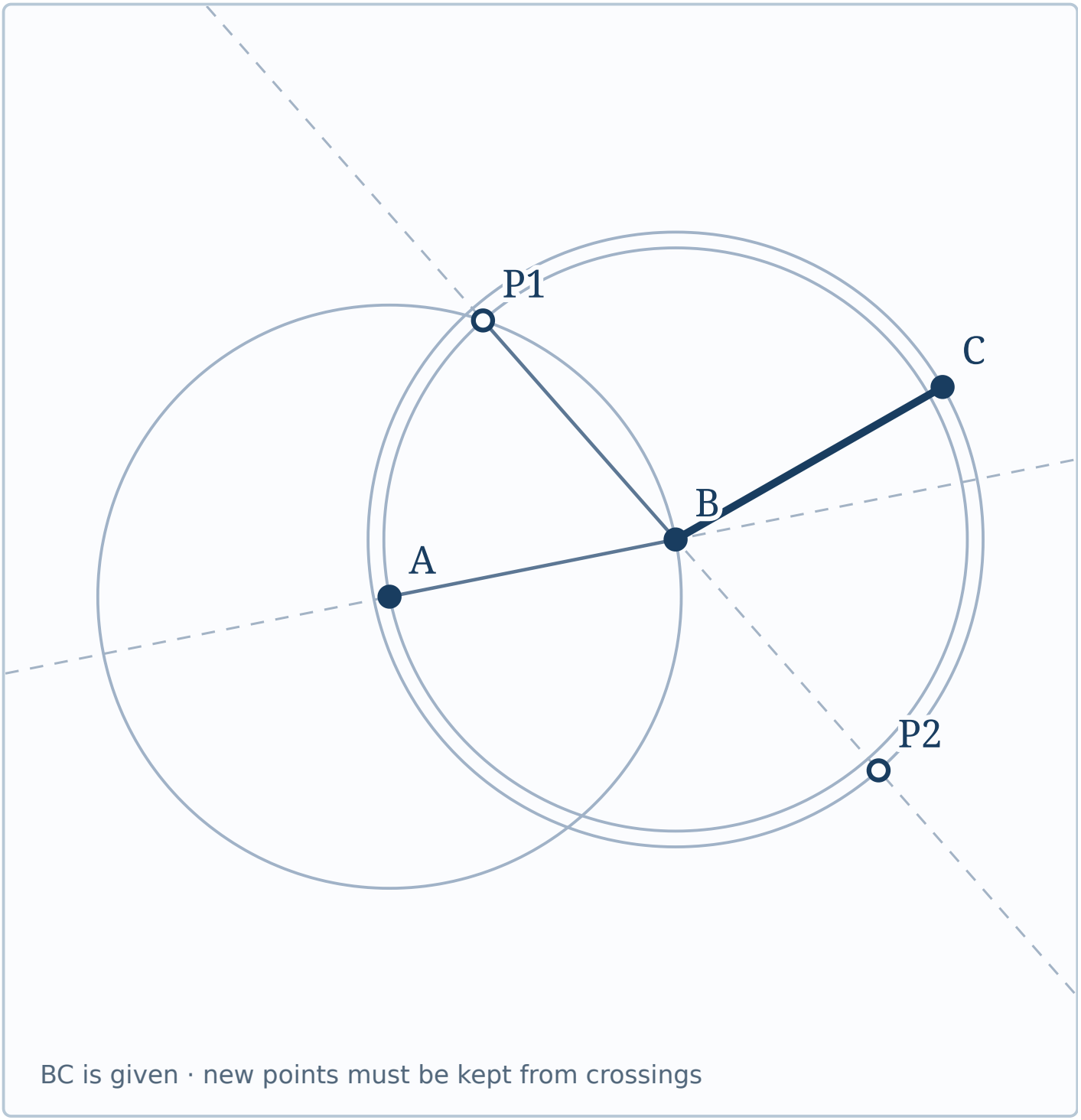


6. Circle centred at B, through C



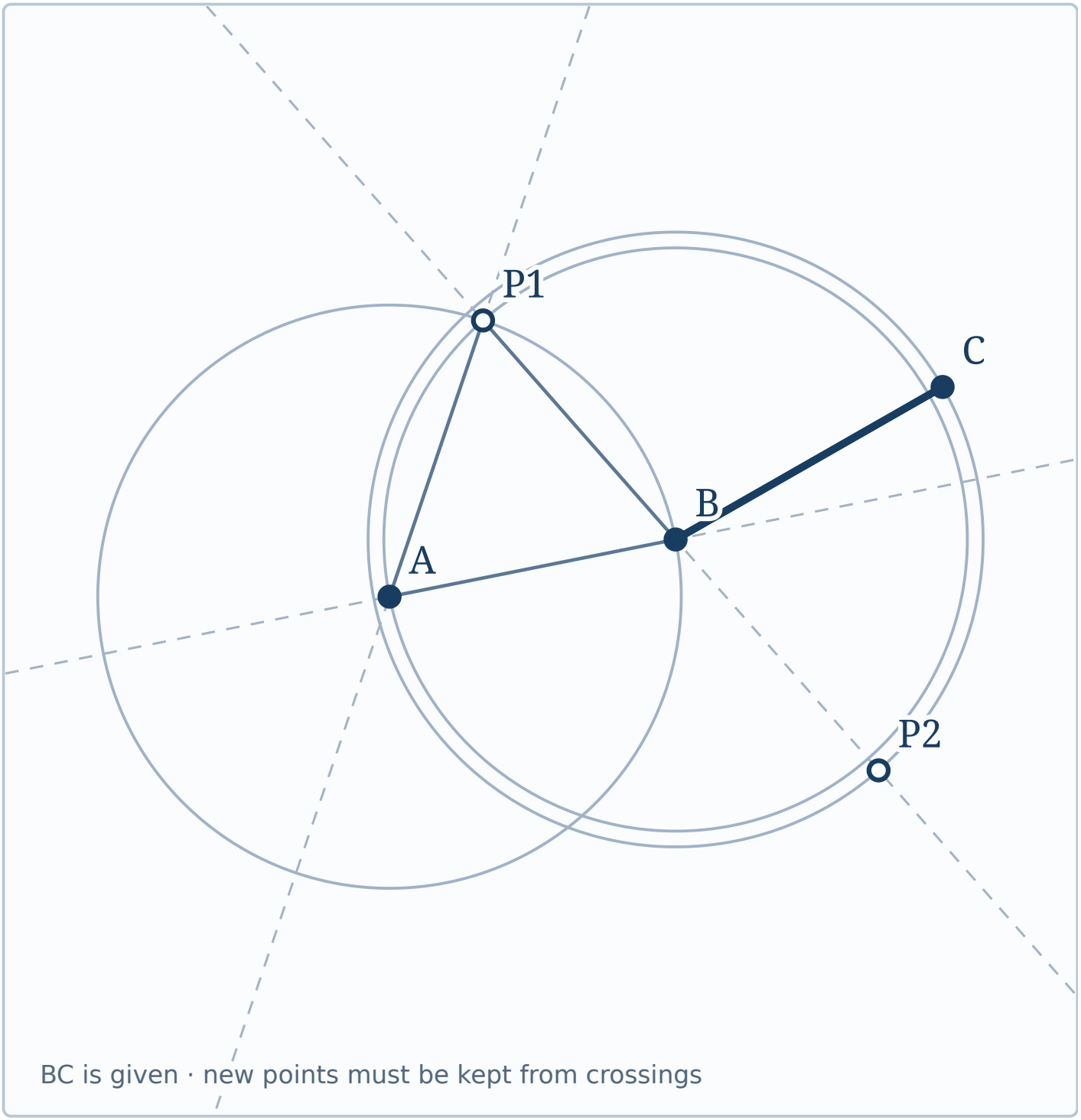
BC is given · new points must be kept from crossings

7. Keep crossing 2 of l4 and c5

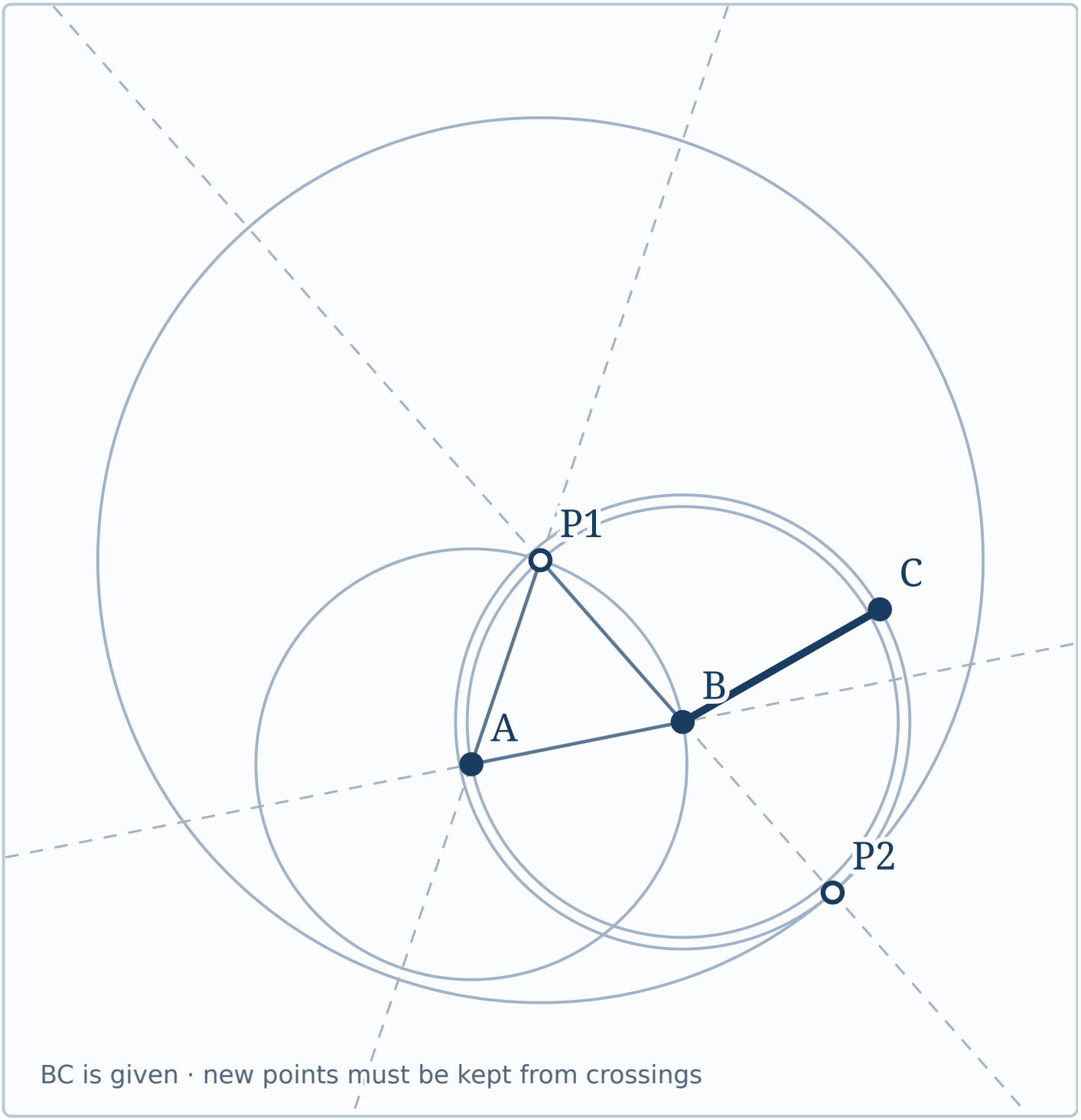


BC is given · new points must be kept from crossings

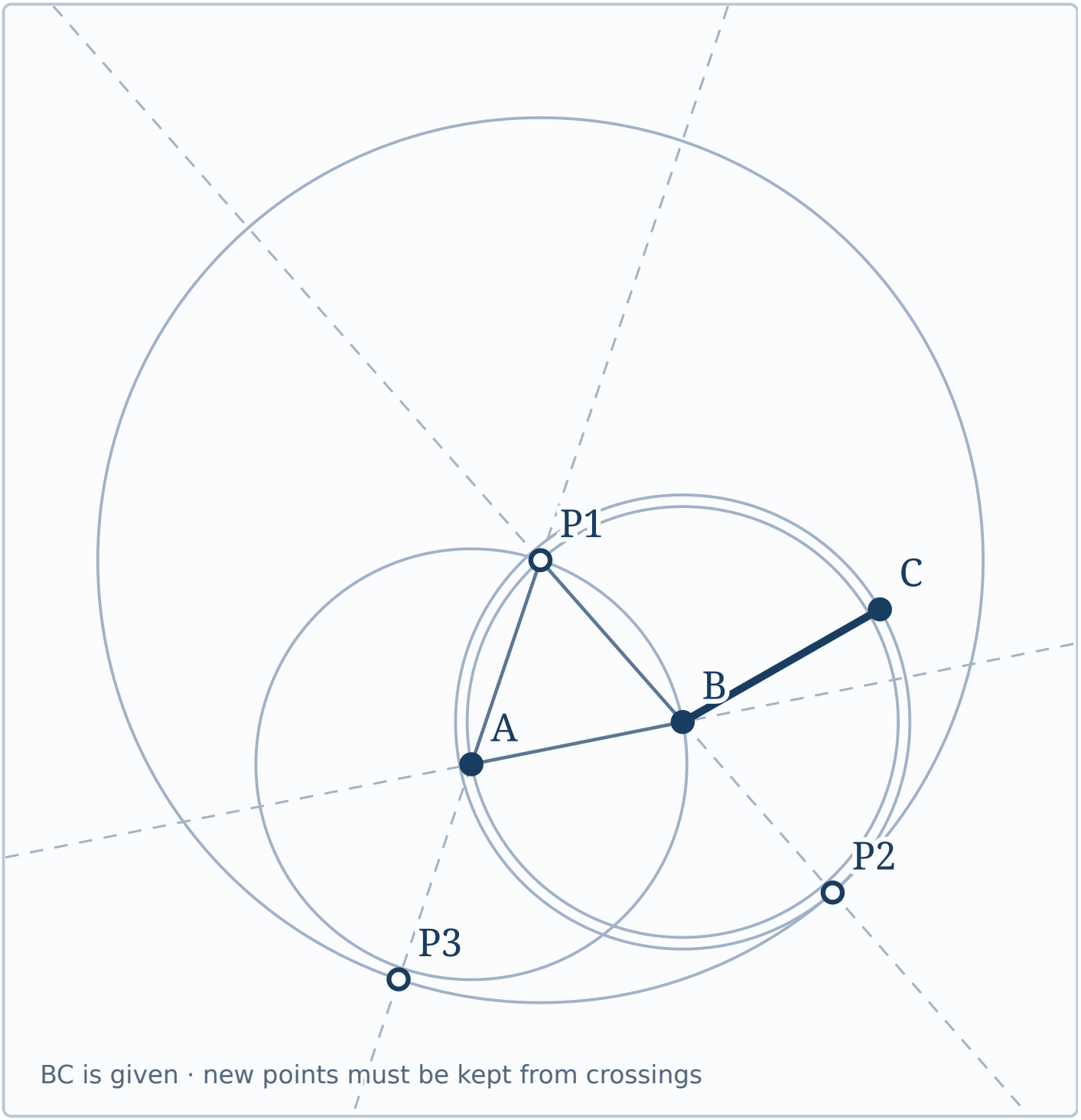
8. Join and continue P1–A



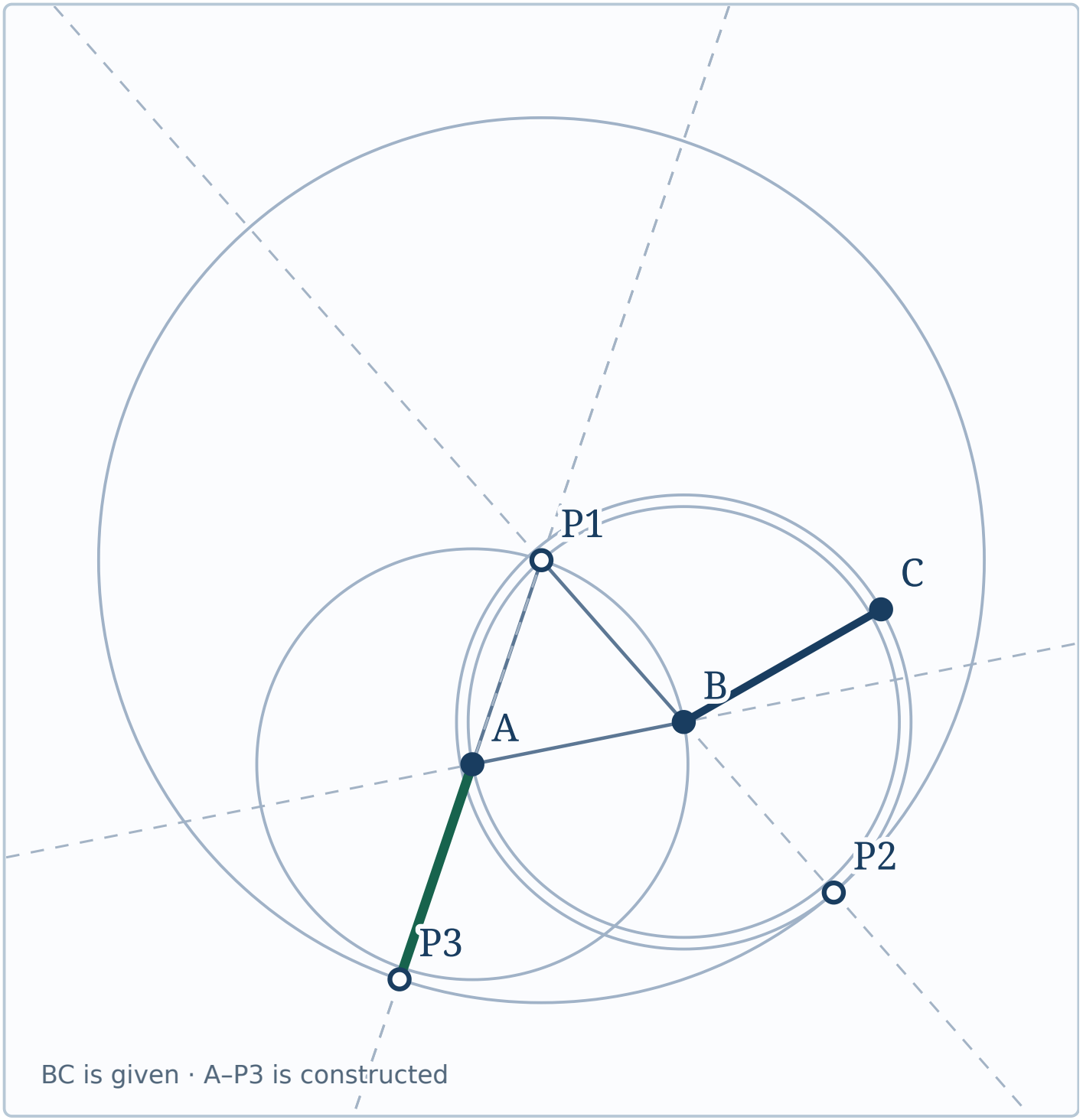
9. Circle centred at P1, through P2



10. Keep crossing 2 of l6 and c7



11. Join and continue A–P3



Read the construction you are making

The complete statement and proof of I.2, in the course’s existing transcription. Your A, B, C correspond to A, B, Γ; new points receive P-numbers in order of creation. A completed explanation supplies the remaining correspondence. Step divisions and language notes are modern teaching aids.

The task

Πρὸς τῷ δοθέντι σημείῳ τῇ δοθείσῃ εὐθείᾳ ἴσην εὐθεῖαν θέσθαι.

Ad datum punctum datae rectae aequalem rectam constituere.

To place, at a given point, a straight line equal to a given straight line.

---

1. A point and a separate length

Ἔστω τὸ μὲν δοθὲν σημεῖον τὸ Α, ἡ δὲ δοθεῖσα εὐθεῖα ἡ ΒΓ· δεῖ δὴ πρὸς τῷ Α σημείῳ τῇ δοθείσῃ εὐθείᾳ τῇ ΒΓ ἴσην εὐθεῖαν θέσθαι.

Sit datum punctum A, data autem recta ΒΓ. oportet igitur ad punctum A datae rectae ΒΓ aequalem rectam constituere.

Let A be the given point and ΒΓ the given straight line. We must place at A a straight line equal to ΒΓ.

μέν … δέ pairs two items: the given point on the one hand, the given line on the other. δοθέν and δοθεῖσα both mean ‘given’; their forms agree with the different nouns.

Greek p.12 · Latin p.13 · Read on the prepared plate

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2. Join A to B

Ἐπεζεύχθω γὰρ ἀπὸ τοῦ Α σημείου ἐπὶ τὸ Β σημεῖον εὐθεῖα ἡ ΑΒ,

ducatur enim a puncto A ad B punctum recta AB,

Draw the straight line AB from point A to point B.

ἐπεζεύχθω is singular: one line is to be joined. Compare the plural ἐπεζεύχθωσαν in I.1, where two lines were joined.

Greek p.12 · Latin p.13 · Read on the prepared plate

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3. Use I.1 on the new line

καὶ συνεστάτω ἐπ᾽ αὐτῆς τρίγωνον ἰσόπλευρον τὸ ΔΑΒ.

et in ea construatur triangulus aequilaterus ΔΑΒ,

On it construct the equilateral triangle ΔAB.

ἐπ᾽ αὐτῆς and in ea both mean ‘on it’, referring back to AB. The point name Δ is delta; this triangle supplies the equality ΔA = ΔB.

Greek p.12 · Latin p.13 · Read on the prepared plate

---

4. Extend past A and B

καὶ ἐκβεβλήσθωσαν ἐπ᾽ εὐθείας ταῖς ΔΑ, ΔΒ εὐθεῖαι αἱ ΑΕ, ΒΖ,

et producantur in directum rectae ΔΑ, ΔΒ, ut fiant ΑΕ, ΒΖ,

Continue ΔA and ΔB in straight lines beyond A and B, forming AE and BZ.

ἐκβεβλήσθωσαν commands that the lines be extended. Latin producantur in directum says ‘let them be extended straight on’. The directions matter: go away from Δ.

Greek p.12, 14 · Latin p.13, 15 · Read on the prepared plate

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5. Mark the reference length from B

καὶ κέντρῳ μὲν τῷ Β διαστήματι δὲ τῷ ΒΓ κύκλος γεγράφθω ὁ ΓΗΘ.

et centro Β radio autem ΒΓ circulus describatur ΓΗΘ,

With centre B and radius BΓ, draw circle ΓHΘ. H lies on the extension beyond B.

H is eta, not the Latin letter H. The circle is named by points on its circumference; Θ is another such point in the source figure. This plate labels the points needed for the argument.

Greek p.14 · Latin p.15 · Read on the prepared plate

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6. Reach both extended sides from Δ

καὶ πάλιν κέντρῳ τῷ Δ καὶ διαστήματι τῷ ΔΗ κύκλος γεγράφθω ὁ ΗΚΛ.

et rursus centro Δ radio autem ΔΗ circulus describatur ΗΚΛ.

Again, with centre Δ and radius ΔH, draw circle ΗΚΛ. Λ lies on the extension beyond A.

Λ is lambda. The names ΗΚΛ identify the second circle. K is not needed in the subtraction argument, so the plate shows the working arc through H and Λ.

Greek p.14 · Latin p.15 · Read on the prepared plate

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7. Identify the first equal radii

Ἐπεὶ οὖν τὸ Β σημεῖον κέντρον ἐστὶ τοῦ ΓΗΘ, ἴση ἐστὶν ἡ ΒΓ τῇ ΒΗ.

iam quoniam Β punctum centrum est circuli ΓΗΘ, erit ΒΓ = ΒΗ.

Since B is the centre of circle ΓΗΘ, ΒΓ equals ΒΗ.

The radius argument from I.1 appears again. ΒΓ is the given length, and BH is now an equal length on the extended side.

Greek p.14 · Latin p.15 · Read on the prepared plate

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8. Identify equal wholes and equal parts

πάλιν, ἐπεὶ τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ ΗΚΛ κύκλου, ἴση ἐστὶν ἡ ΔΛ τῇ ΔΗ, ὧν ἡ ΔΑ τῇ ΔΒ ἴση ἐστίν.

rursus quoniam Δ punctum centrum est circuli ΗΚΛ, erit ΔΛ = ΔΗ, quarum partes ΔΑ, ΔΒ aequales.

Since Δ is the centre of circle ΗΚΛ, ΔΛ equals ΔΗ; within them, the parts ΔA and ΔB are equal.

ὧν means ‘of which’. Heiberg’s Latin makes the implied relationship explicit with quarum partes: ‘whose parts’. The equal parts come from the equilateral triangle.

Greek p.14 · Latin p.15 · Read on the prepared plate

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9. Subtract the equal parts

λοιπὴ ἄρα ἡ ΑΛ λοιπῇ τῇ ΒΗ ἐστιν ἴση.

itaque ΑΛ = ΒΗ.

Therefore the remainder ΑΛ is equal to the remainder ΒΗ.

λοιπή means ‘remaining’. Greek names the remainders, while the Latin prints an equation. The language encodes an operation: remove ΔΑ and ΔΒ, leaving ΑΛ and ΒΗ.

Greek p.14 · Latin p.15 · Read on the prepared plate

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10. Connect the remainder to the reference

ἐδείχθη δὲ καὶ ἡ ΒΓ τῇ ΒΗ ἴση· ἑκατέρα ἄρα τῶν ΑΛ, ΒΓ τῇ ΒΗ ἐστιν ἴση. τὰ δὲ τῷ αὐτῷ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα· καὶ ἡ ΑΛ ἄρα τῇ ΒΓ ἐστιν ἴση.

sed demonstratum est ΒΓ = ΒΗ. itaque utraque ΑΛ, ΒΓ rectae ΒΗ aequalis est. uerum quae eidem aequalia sunt, etiam inter se aequalia sunt. ergo etiam ΑΛ = ΒΓ.

ΒΓ was also shown equal to ΒΗ. Since ΑΛ and ΒΓ are each equal to ΒΗ, they are equal to one another: ΑΛ = ΒΓ.

ἑκατέρα means ‘each of the two’; Latin utraque has the same role. The proof repeats common notion 1 rather than relying on the appearance of the diagram.

Greek p.14 · Latin p.15 · Read on the prepared plate

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11. The new length starts at A

Πρὸς ἄρα τῷ δοθέντι σημείῳ τῷ Α τῇ δοθείσῃ εὐθείᾳ τῇ ΒΓ ἴση εὐθεῖα κεῖται ἡ ΑΛ· ὅπερ ἔδει ποιῆσαι.

Ergo ad datum punctum A datae rectae BΓ aequalis constituta est recta AΛ; quod oportebat fieri.

At the given point A, the straight line AΛ has been placed equal to the given straight line BΓ: which was to be done.

κεῖται (‘lies’ or ‘has been placed’) answers the opening task θέσθαι (‘to place’). Latin constituere is answered by constituta est. The end echoes the beginning.

Greek p.14 · Latin p.15 · Read on the prepared plate

Tools, text and limitations

A circle has an existing centre and passes through an existing point. The straight-line tool combines joining two points and continuing the join in both directions (postulates 1 and 2). New points must come from isolated crossings. These restrictions are a modern teaching interface, not a reconstruction of an ancient physical compass.

The desk recognizes the I.2 equal-remainder dependency chain, with either equilateral vertex or reference anchor. It does not establish every possible valid solution. Screen measurements are approximate and never certify a proof. Intersections use floating-point real-plane coordinates; tangency and numerically unresolved intersections are deliberately not kept. Euclid’s stated postulates do not themselves supply a complete continuity justification for circle intersections.

Greek and Heiberg’s Latin: *Euclidis opera omnia*, vol. I (Leipzig: Teubner, 1883), pp.12–15. University of Toronto scan. Full pages retain the apparatus, which is not transcribed. The course normalizes ϑ to θ, removes line wrapping and inline Latin reference marks, and keeps Greek labels in both languages. English and notes are new.

This reading embeds four complete page images, the Greek font, all replay diagrams and editable JSON. It runs no scripts and needs no network for reading. Links to the app and source volume require a connection.

Heiberg 1883, printed p.12 · Greek · SHA-256
e5cc7d6bf3a2ec8dc3bcbf88803b8ee9e3d0315ae29221ed7d2fe9768356c2fd

γεγράφθω ὁ ΒΓΔ, καὶ πάλιν κέντρῳ μὲν τῷ Β διαστήματι δὲ τῷ ΒΑ κύκλος γεγράφθω ὁ ΑΓΕ, καὶ ἀπὸ τοῦ Γ σημείου, καθ' ὃ τέμνουσιν ἀλλήλους οἱ κύκλοι, ἐπὶ τὰ Α, Β σημεῖα ἐπεζεύχθωσαν εὐθεῖαι αἱ ΓΑ, ΓΒ.

Καὶ ἐπεὶ τὸ Α σημεῖον κέντρον ἐστὶ τοῦ ΓΔΒ κύκλου, ἴση ἐστὶν ἡ ΑΓ τῇ ΑΒ· πάλιν, ἐπεὶ τὸ Β σημεῖον κέντρον ἐστὶ τοῦ ΓΑΕ κύκλου, ἴση ἐστὶν ἡ ΒΓ τῇ ΒΑ. ἐδείχθη δὲ καὶ ἡ ΓΑ τῇ ΑΒ ἴση· ἑκατέρα ἄρα τῶν ΓΑ, ΓΒ τῇ ΑΒ ἐστιν ἴση. τὰ δὲ τῷ αὐτῷ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα· καὶ ἡ ΓΑ ἄρα τῇ ΓΒ ἐστιν ἴση· αἱ τρεῖς ἄρα αἱ ΓΑ, ΑΒ, ΒΓ ἴσαι ἀλλήλαις εἰσίν.

ἰσόπλευρον ἄρα ἐστὶ τὸ ΑΒΓ τρίγωνον. καὶ συνέσταται ἐπὶ τῆς δοθείσης εὐθείας πεπερασμένης τῆς ΑΒ.

[Ἐπὶ τῆς δοθείσης ἄρα εὐθείας πεπερασμένης τρίγωνον ἰσόπλευρον συνέσταται] ὅπερ ἔδει ποιῆσαι.

β΄.

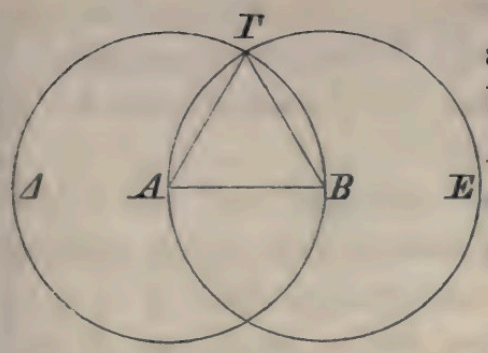
Πρὸς τῷ δοθέντι σημείῳ τῇ δοθείσῃ εὐθείᾳ ἴσην εὐθεῖαν θέσθαι.

Ἔστω τὸ μὲν δοθὲν σημεῖον τὸ Α, ἡ δὲ δοθεῖσα εὐθεῖα ἡ ΒΓ· δεῖ δὴ πρὸς τῷ Α σημείῳ τῇ δοθείσῃ εὐθείᾳ τῇ ΒΓ ἴσην εὐθεῖαν θέσθαι.

Ἐπεζεύχθω γὰρ ἀπὸ τοῦ Α σημείου ἐπὶ τὸ Β σημεῖον εὐθεῖα ἡ ΑΒ, καὶ συνεστάτω ἐπ' αὐτῆς τρίγωνον ἰσόπλευρον τὸ ΔΑΒ, καὶ ἐκβεβλήσθωσαν ἐπ'

II. Archimedes I p. 14, 1. Boetius p. 380, 3 [p. 391].

1. ΒΓΔ] P, V m. 1; ΓΔΒ F b p, V e corr.; ΓΒΔ in ras. B. μέν] om. b. τῷ] τό φ. 2. ΑΓΕ] P, V m. 1; ΓΑΕ B F b p, V e corr. 6. Post Α ras. 10 litt. b. ἐστίν P. ΓΔΒ] Δ in



et rursus centro B radio autem BA circulus describatur $A\Gamma E$, et a puncto Γ, in quo circuli inter se secant, ad puncta A, B ducantur rectae ΓA, ΓB. iam quoniam punctum A centrum est circuli $\Gamma\Delta B$, erit $A\Gamma = AB$. rursus quoniam B punctum centrum est circuli ΓAE, est $B\Gamma = BA$. sed demonstratum est etiam $\Gamma A = AB$. quare utraque ΓA, ΓB rectae AB aequalis est. quae autem eidem aequalia sunt, etiam inter se aequalia sunt [*κ. ἔνν.* 1]. itaque etiam $\Gamma A = \Gamma B$. itaque ΓA, AB, $B\Gamma$ aequales sunt. quare triangulus $AB\Gamma$ aequilaterus est; et in data recta terminata AB constructus est. quod oportebat fieri.

II.

Ad datum punctum datae rectae aequalem rectam constituere.

Sit datum punctum A, data autem recta $B\Gamma$. oportet igitur ad punctum A datae rectae $B\Gamma$ aequalem rectam constituere.

ducatur enim a puncto A ad B punctum recta AB [*αἴτ.* 1], et in ea construatur triangulus aequilaterus ΔAB [prop. I], et producantur in directum rectae

ras. est in V, ΔB in B; $B\Gamma\Delta$ P. 7. *ἐστιν ἴση* BF. 8. *ἐστίν* P. ΓAE] in ras. B, $A\Gamma E$ P. 12. *ἴση ἐστίν* V. AB] ΓB *φ*. 14. *ἐστίν* P. *συνίσταται* PBV (in b non liquet). 16. *ἐπὶ τῆς* — 17. *συνέσταται* om. codd. omnes; e Proclo solo p. 210 recepit August; uix genuina sunt. 22. *τῇ δοθείσῃ εὐθείᾳ*] P; om. Theon (BFVpb). 23. $B\Gamma$ *εὐθεία* V. 24. *γάρ*] om. F. 26. ΔAB] eras. F. Ante *ἐκβεβλ.* in V add. supra: *προσ-*.

Heiberg 1883, printed p.14 · Greek · SHA-256
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εὐθείας ταῖς ΔΑ, ΔΒ εὐθεῖαι αἱ ΑΕ, ΒΖ, καὶ κέντρῳ μὲν τῷ Β διαστήματι δὲ τῷ ΒΓ κύκλος γεγράφθω ὁ ΓΗΘ, καὶ πάλιν κέντρῳ τῷ Δ καὶ διαστήματι τῷ ΔΗ κύκλος γεγράφθω ὁ ΗΚΛ.

Ἐπεὶ οὖν τὸ Β σημεῖον κέντρον ἐστὶ τοῦ ΓΗΘ, ἴση ἐστὶν ἡ ΒΓ τῇ ΒΗ. πάλιν, ἐπεὶ τὸ Δ σημεῖον κέντρον ἐστὶ τοῦ ΗΚΛ κύκλου, ἴση ἐστὶν ἡ ΔΛ τῇ ΔΗ, ὧν ἡ ΔΑ τῇ ΔΒ ἴση ἐστίν. λοιπὴ ἄρα ἡ ΑΛ λοιπῇ τῇ ΒΗ ἐστιν ἴση. ἐδείχθη δὲ καὶ ἡ ΒΓ τῇ ΒΗ ἴση· ἑκατέρα ἄρα τῶν ΑΛ, ΒΓ τῇ ΒΗ ἐστιν ἴση. τὰ δὲ τῷ αὐτῷ ἴσα καὶ ἀλλήλοις ἐστὶν ἴσα· καὶ ἡ ΑΛ ἄρα τῇ ΒΓ ἐστιν ἴση.

Πρὸς ἄρα τῷ δοθέντι σημείῳ τῷ Α τῇ δοθείσῃ εὐθείᾳ τῇ ΒΓ ἴση εὐθεῖα κεῖται ἡ ΑΛ· ὅπερ ἔδει ποιῆσαι.

γ΄.

Δύο δοθεισῶν εὐθειῶν ἀνίσων ἀπὸ τῆς μείζονος τῇ ἐλάσσονι ἴσην εὐθεῖαν ἀφελεῖν.

Ἔστωσαν αἱ δοθεῖσαι δύο εὐθεῖαι ἄνισοι αἱ ΑΒ, Γ, ὧν μείζων ἔστω ἡ ΑΒ· δεῖ δὴ ἀπὸ τῆς μείζονος τῆς ΑΒ τῇ ἐλάσσονι τῇ Γ ἴσην εὐθεῖαν ἀφελεῖν.

Κείσθω πρὸς τῷ Α σημείῳ τῇ Γ εὐθείᾳ ἴση ἡ ΑΔ· καὶ κέντρῳ μὲν τῷ Α διαστήματι δὲ τῷ ΑΔ κύκλος γεγράφθω ὁ ΔΕΖ.

III. Boetius p. 380, 5 [p. 392].

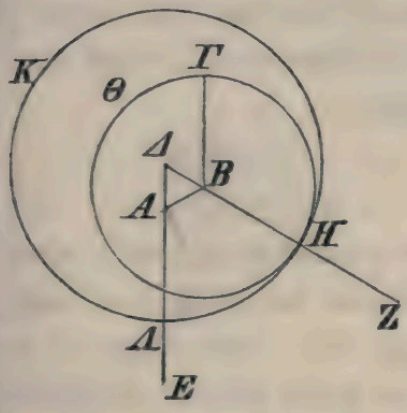
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1. εὐθείας FV. 3. κέντρῳ μέν V. τῷ] bis B (in fine et initio linn.). καὶ διαστήματι] διαστήματι δέ V. 5. ΓΗΘ κύκλου BFV, P m. rec. 6. ΒΓ] ΓΒ F. καὶ πάλιν V; πάλιν δέ (supra) p. 7. ἐστίν P. 8. ἐστίν] PF; ἐστι uulgo. 9. τῇ] om. b. 10. τῇ ΒΗ] (alt.) supra b. 11. ἴσα] (alt.) -α in ras. P. 12. ΒΓ] ΓΒ F. 13. Ante πρός ras. unius litt. b. 18. ἐλάττονι BF. εὐθεῖαν] om. Proclus. 19. δύο] om. F. ἄνισοι] ἀν- supra m. 1 F. 20. Post Γ ras. 1 litt.

Heiberg 1883, printed p.15 · Latin · SHA-256
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ΔA, ΔB, ut fiant AE, BZ, et centro B radio autem $B\Gamma$ circulus describatur [*αἴτ.* 2] $\Gamma H\Theta$, et rursus centro Δ radio autem ΔH circulus describatur $HK\Lambda$.

iam quoniam B punctum centrum est circuli $\Gamma H\Theta$,



erit $B\Gamma = BH$. rursus quoniam Δ punctum centrum est circuli $HK\Lambda$, erit

$$\Delta\Lambda = \Delta H,$$

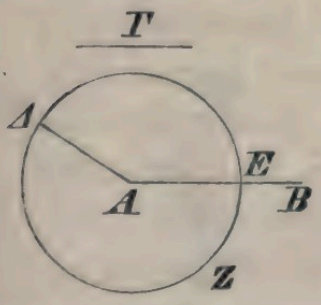
quarum partes ΔA, ΔB aequales. itaque $A\Lambda = BH$ [*κ. ἔννν.* 3]. sed demonstratum est $B\Gamma = BH$. itaque utraque $A\Lambda$, $B\Gamma$ rectae BH aequalis est. uerum quae eidem aequalia sunt, etiam inter se aequalia sunt [*κ. ἔννν.* 1]. ergo etiam $A\Lambda = B\Gamma$.

Ergo ad datum punctum A datae rectae $B\Gamma$ aequalis constituta est recta $A\Lambda$; quod oportebat fieri.

III.

Datis duabus rectis inaequalibus rectam minori aequalem a maiore abscindere.

Sint duae datae rectae inaequales AB, Γ, quarum maior sit AB. oportet igitur a maiore AB minori Γ aequalem rectam abscindere. constituatur



ad A punctum rectae Γ aequalis $A\Delta$ [propr. II], et centro A radio autem $A\Delta$ describatur circulus ΔEZ [*αἴτ.* 2].

P, ut lin. 21. 22. 22. Post *κείσθω* in P supra scr. m. 1 *γάρ*, idem V mg. 23. $A\Delta$] (alt.) in ras. V; utrumque corr. ex AE P m. rec. 24. ΔEZ] ex EZI P m. rec.; $ZE\Delta$ B.